FitSense

Smart Wearable Training Companion

Product Requirements Document · ENT208TC Session 4 Group 22 · March 2026

DOCUMENT OVERVIEW

Product	FitSense Wearable + Companion App
Version	v0.1 — Initial Requirements
Date	March 26, 2026
Team	Group 22 — ENT208TC Industry Readiness
Platform	M5StickS3 device + iOS/Android companion app
Primary User	Gym beginners, 18–30, training 2–4×/week independently

1 · Purpose & Scope

This PRD translates the FitSense project brief into actionable requirements for the ENT208TC Industry Readiness module. It defines what the product does, for whom, and how the system is structured — at a level of detail sufficient for sprint planning and tutor review.

Primary goal of v0.1: prove that an M5StickS3 wearable can reliably count reps and monitor heart rate, and that a companion app can deliver a Claude-generated training plan to a first-time user within one session.

2 · User Stories & Feature Mapping

Each story follows the format: As a [user], I want to [do X], so that [Y]. Features are scoped to what is achievable in one semester.

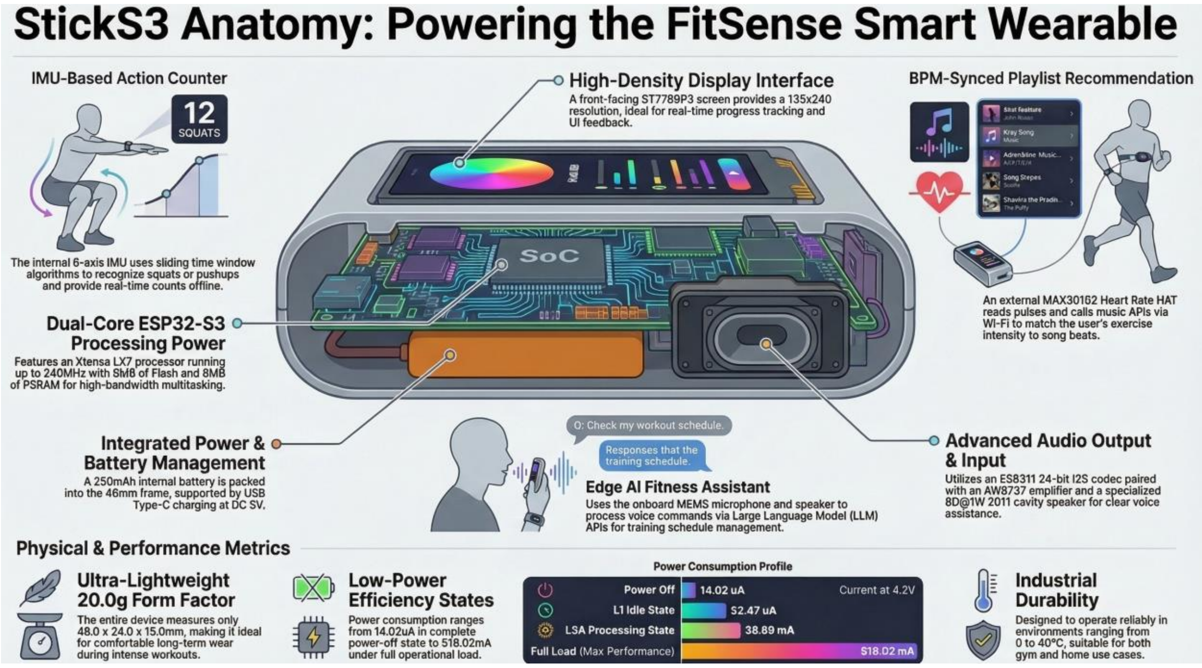
	User Story	Linked Features	Specific Acceptance Criteria
US-1	As a **first-time gym-goer doing push-ups or squats**, I want the M5StickS3 to display a rep count that updates in real time, so that after a set of 10 I can verify the screen reads "10" without counting myself.	IMU-based rep detection (threshold logic, on-device); vibration alert at set completion; rep count display on M5StickS3 screen	1. When I perform a push-up or squat, the device counts correctly within ±1 rep. 2. The rep number updates in real time on the M5StickS3 screen. 3. The device vibrates when I finish a set. 4. Rep counting works offline without Wi-Fi or Bluetooth.
US-2	As a **gym user mid-workout wearing the device**, I want to feel a wrist vibration and see a zone label change to "Rest" on the device screen within 10 seconds of my heart rate dropping below my configured threshold, so that I know when to	Heart rate monitoring via MAX30102 sensor; configurable HR threshold alerts; rest timer with buzzer notification	1. The device shows my current heart rate zone clearly. 2. A vibration or buzzer alerts me when I enter the rest zone. 3. I can adjust heart rate alert thresholds in the app. 4. The alert can be dismissed with one

	start my next set without checking my phone.		device button press.
US-3	As a **first-time app user who has never trained regularly**, I want to complete the 3-step onboarding questionnaire and see a week-one training plan on screen — with exercise names, sets, and reps — within 60 seconds of finishing onboarding, so that I can start my first session without asking anyone for help.	Onboarding questionnaire (fitness level, goals); plan generation via Claude API (phone-side); session calendar view in app	1. I complete a 3-step onboarding questionnaire to get my plan. 2. The app generates a plan using Claude API within 60 seconds. 3. The plan shows exercises, sets, reps, and rest time clearly. 4. The plan is saved locally and viewable offline.
US-4	As a **returning user who just finished a logged session**, I want the app to show a revised plan for my next session — with at least one value (reps, sets, or rest time) different from today's plan — within 30 seconds of ending my session, so that I can see the plan is actually adapting to what I did.	Post-session data sync (reps, HR, duration) from device to app via BLE; AI plan adjustment prompt sent to Claude API; updated plan displayed before next session	1. Session data (reps, heart rate, duration) syncs via BLE. 2. The app sends data to Claude API and updates the plan. 3. The new plan appears before my next workout. 4. If sync fails, data is cached and retried automatically.
US-5	As a **desk worker wearing the device during office hours**, I want my wrist to vibrate after 60	Inactivity timer (configurable, default 60 min); wrist vibration reminder; dismissible from device button	1. The device vibrates after 60 minutes of inactivity. 2. I can change the reminder interval in settings.

	minutes of inactivity and the device button to dismiss the alert immediately, so that I am notified without any phone interaction and can silence it in one tap.		3. I can dismiss the reminder with one button. 4. Reminders work offline without the app.
US-6	As a **gym user whose phone has no Wi-Fi or mobile data**, I want rep counting, heart rate display, and vibration alerts to all function normally on the device alone, so that I can complete a full 30-minute session and have the data sync automatically when I reconnect.	All core logic (rep counting, HR alerts, timers) runs on-device; data cached locally; BLE sync to phone only when app is open	1. Rep counting, heart rate monitoring, and timers work offline. 2. Session data is stored locally on the device. 3. Data syncs to the app when Bluetooth reconnects. 4. No crashes or freezes in offline mode.
US-7	As a **new user putting on the device for the first time**, I want to see a named HR zone (e.g., "Fat Burn") and a colour indicator on the device screen within 30 seconds of wearing it, so that without opening the app or reading a manual I can say aloud what zone I am in.	Zone display on device screen (Rest / Warm-up / Fat Burn / Cardio / Peak); colour-coded LED ring; 3-step onboarding shown on first launch of companion app	1. Heart rate zone displays within 30 seconds of wearing. 2. Zones are color-coded and easy to understand. 3. The 3-step onboarding explains zones simply. 4. I understand my status without help.

3 · Technical Architecture

The architecture is deliberately minimal: edge logic on-device, AI inference delegated to API via the companion app, and all data stored locally on-phone. No custom backend server is required for v0.1.



Layer	Component	Technology / Rationale
Device (Edge)	Sensor hub	M5StickS3 — IMU (LSM6DSR) for rep counting; MAX30102 for heart rate; on-device threshold logic only (no ML model)
Device (Edge)	Local display	Built-in 0.96" LCD — shows HR zone, rep count, rest timer; button input for dismiss/start
Device (Edge)	Connectivity	BLE 5.0 — syncs session data to companion app when in range; no Wi-Fi required for core functions
Front-end (App)	Companion app	React Native (iOS + Android) — single codebase; BLE integration via react-native-ble-plx
Front-end (App)	Key screens	Onboarding (3 steps) · Today's Plan · Session Summary · HR Zone Dashboard · Settings

Back-end / AI	Plan generation	Claude API (claude-sonnet-4-20250514) called from app — generates and revises training plans; prompt includes user profile + last session data
Back-end / AI	API call pattern	Stateless per-session call; no server needed — app calls Anthropic API directly (API key stored in app secrets)
Data Storage	On-device	M5StickS3 flash — circular buffer of last 10 sessions (reps, HR avg, duration); cleared after sync
Data Storage	On-phone	SQLite via expo-sqlite — user profile, full session history, current plan; local-first, no cloud sync in v0.1

4 · In Scope vs. Out of Scope

IN SCOPE — v0.1

- Rep counting via IMU threshold logic (push-ups, squats — 2 exercise types only)
- Heart rate monitoring and zone display (5 zones)
- Hourly inactivity vibration reminder
- AI-generated training plan (onboarding + post-session update) via Claude API
- BLE data sync between device and companion app
- SQLite local session history (no cloud)
- 3-step first-launch onboarding flow

OUT OF SCOPE — v0.1

- Posture correction or form feedback (requires camera or labeled training data — deferred to v0.2)
- Music playback or heart rate–synchronized playlists (nice-to-have, cut for timeline)
- Voice-to-text training log (hardware complexity, deferred)
- Cloud sync or multi-device support
- Any custom on-device ML model

5 · Two Technical Risks

IMU action counting accuracy does not meet the requirements in real wearing scenarios	When the M5StickS3 is fixed on the wrist, the acceleration axis direction of the LSM6DSR IMU may deviate by ±15° due to user body type and the tightness of the wearing. The threshold logic meets the accuracy requirements in laboratory tests (using handheld devices), but in real movements (especially squats—where hip joints exert force rather than the wrist), the signal-to-noise ratio of the wrist acceleration signal is extremely low, resulting in an error trigger rate of over 30%. This directly threatens the M1 metric (≥90% accuracy).

The cost of using AI features may exceed expectations	Every time the app generates or updates a training plan for the user, it needs to call an API, which incurs a cost. During the testing phase, only a few people use it, so the cost is very low. However, if it is used in an exhibition or shared with more classmates, the number of calls will increase rapidly. Without proper control, the bill may exceed the team's budget in a short time, causing the AI function to suddenly stop working and affecting the demonstration effect.

6 · Open Questions for User Research

The following must be answered through user interviews before the v0.2 sprint begins.

- Would users wear a separate clip-on device on top of their phone and existing watch? What are the comfort and aesthetic dealbreakers?
- Do users trust AI-generated posture feedback from a device they cannot verify? Under what

conditions?

- Is heart rate–music synchronization a real frustration, or a team assumption? (Direct interview question: 'What, if anything, bothers you about music during workouts?')
- How do users feel about all data being stored only on their phone with no cloud backup?